

“Higher Time Derivatives:
The Good, the Bad & the Ugly”

Virtual Higher Time Derivative Seminar

October 6, 2026

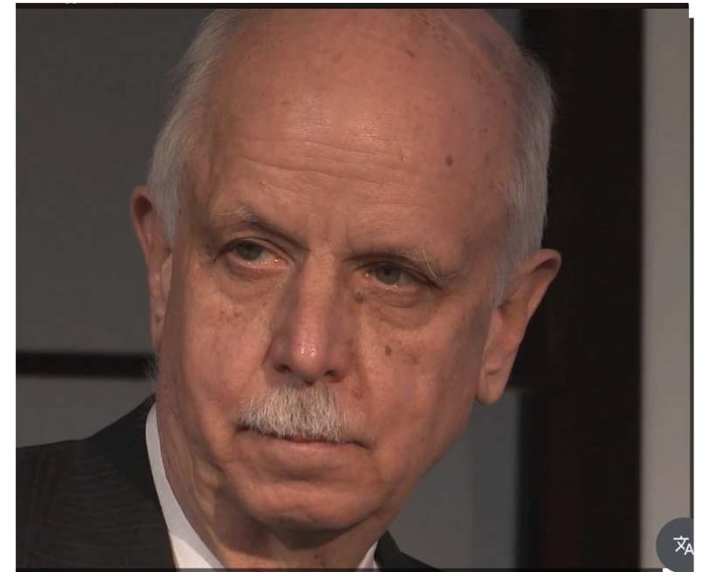
Richard Woodard (U. Florida)

The Good

Stelle's Theorem: PRD 16 (1977) 953-969
(3047 citations on INSPIRE)

- $\mathcal{L} = \frac{(R-2\Lambda)\sqrt{-g}}{16\pi G} + \alpha R^2 \sqrt{-g} + \beta C^{\mu\nu\rho\sigma} C_{\mu\nu\rho\sigma} \sqrt{-g}$
 - is perturbatively renormalizable to all orders
- The R^2 term is certainly ok
 - It's the basis of Starobinsky Inflation!
- The C^2 term is NOT ok
 - This is why there is a “problem of quantum gravity”
- NB: believing the C^2 term is ok (as KS did not) amounts to claiming you have solved the problem of QG!

Kellogg S. Stelle 1948-2025
(My “brother”)



The Bad

Ostrogradsky's Theorem: *Mem.Acad.St.Petersbourg* 6
(1850) 4, 385-517 (somewhat before INSPIRE)

- Eqns of motion with HTD's require more IVD
 - Hence, more canonical Q 's and P 's
- Hamiltonian is LINEAR in the extra P 's
- Triggers a virulent, kinetic instability if:
 - Theory is “interacting” (i.e., nonlinear eqns)
 - Has continuum DoF's (i.e., a field theory)
- Why Newton was right about $F = ma$
 - Strongest constraint on fundamental theory

Mikhail Ostrogradsky
(1801-1862)



How Lower Derivatives Work (for point particles)

- Dynamical variable $q(t)$ & Lagrangian $L(q, \dot{q})$
 - $\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) = \frac{\partial L}{\partial q}$ initial conditions $\rightarrow$ 2 canonical variables
- Canonical formulation
 - $Q = q$ & $P = \frac{\partial L}{\partial \dot{q}} \rightarrow \dot{q} = v(Q, P)$ (nondegeneracy)
 - $H(Q, P) = P v(Q, P) - L(Q, v(Q, P))$
- Hamilton's equations generate time evolution
 - $\dot{Q} = \frac{\partial H}{\partial P} = v + P \frac{\partial v}{\partial P} - \frac{\partial L}{\partial \dot{q}} \frac{\partial v}{\partial P} = v$ 
 - $\dot{P} = -\frac{\partial H}{\partial Q} = -P \frac{\partial v}{\partial Q} + \frac{\partial L}{\partial q} + \frac{\partial L}{\partial \dot{q}} \frac{\partial v}{\partial Q} = \frac{\partial L}{\partial q}$ 
- $H(Q, P)$ can be bounded below

Higher Derivatives (Ostrogradsky 1850)

- Lagrangian $L(q, \dot{q}, \ddot{q})$
 - $\frac{d}{dt} \left[\frac{\partial L}{\partial \dot{q}} - \frac{d}{dt} \left(\frac{\partial L}{\partial \ddot{q}} \right) \right] = \frac{\partial L}{\partial q}$ 4 initial conditions $\rightarrow$ 4 canonical coordinates
- Canonical Formulation
 - $Q_1 = q$, $P_1 = \frac{\partial L}{\partial \dot{q}} - \frac{d}{dt} \left(\frac{\partial L}{\partial \ddot{q}} \right)$, $Q_2 = \dot{q}$, $P_2 = \frac{\partial L}{\partial \ddot{q}} \rightarrow \ddot{q} = a(\vec{Q}, P_2)$ (ND)
 - $H(\vec{Q}, \vec{P}) = P_1 Q_2 + P_2 a(\vec{Q}, P_2) - L(Q_1, Q_2, a(\vec{Q}, P_2))$
- Hamilton's equations generate time evolution
 - $\dot{Q}_1 = \frac{\partial H}{\partial P_1} = Q_2$ 😊, $\dot{Q}_2 = \frac{\partial H}{\partial P_2} = a + P_2 \frac{\partial a}{\partial P_2} - \frac{\partial L}{\partial \ddot{q}} \frac{\partial a}{\partial P_2} = a$ 😊
 - $\dot{P}_2 = -\frac{\partial H}{\partial Q_2} = -P_1 - P_2 \frac{\partial a}{\partial Q_2} + \frac{\partial L}{\partial \dot{q}} + \frac{\partial L}{\partial \ddot{q}} \frac{\partial a}{\partial Q_2} = \frac{\partial L}{\partial \dot{q}} - P_1$ 😊
 - $\dot{P}_1 = -\frac{\partial H}{\partial Q_1} = -P_2 \frac{\partial a}{\partial Q_1} + \frac{\partial L}{\partial q} + \frac{\partial L}{\partial \ddot{q}} \frac{\partial a}{\partial Q_1} = \frac{\partial L}{\partial q}$ 😊
- H is **linear** in $P_1 \rightarrow$ not bounded below (or above) ⚡

Extensions

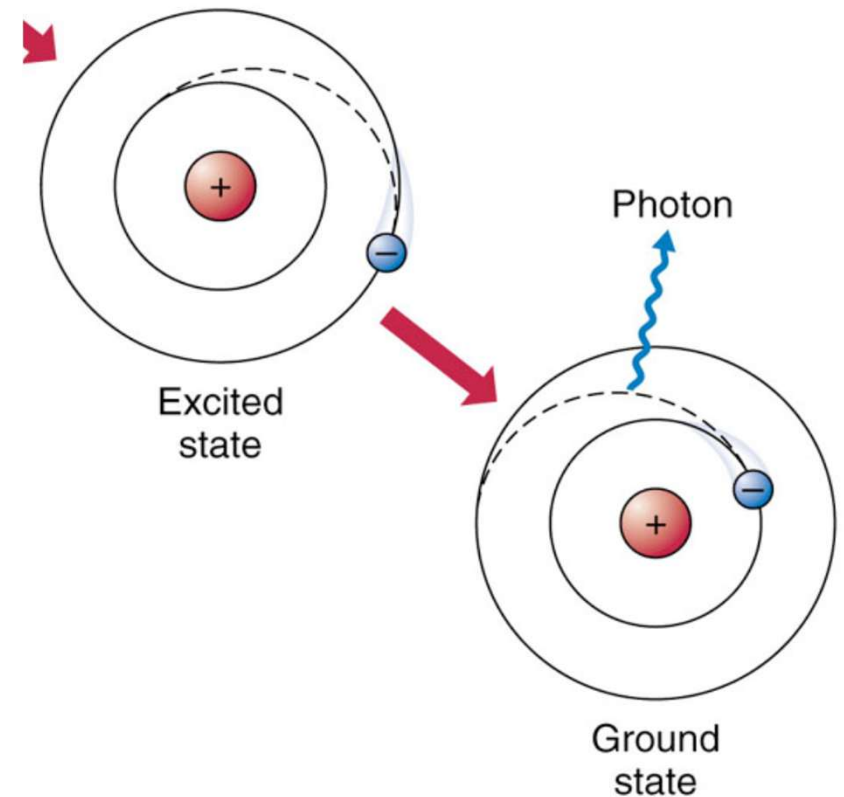
- Even more derivatives $\rightarrow$ problem just gets worse
 - $L\left(q, \dot{q}, \dots, \frac{d^N q}{dt^N}\right) \rightarrow H = Q_2 P_1 + Q_3 P_2 + \dots + Q_N P_{N-1} - f(\vec{Q}, P_N)$
 - Linear in all but one momenta
- Field theories $\rightarrow$ only time derivatives matter
 - $\mathcal{L}(\varphi, \dot{\varphi}, \ddot{\varphi}) \rightarrow H = \int d^3x \{ \Phi_2(\vec{x}) \Pi_1(\vec{x}) - f(\Phi_1, \Phi_2, \Pi_2) \}$
 - Linear canonical direction for each space point $\vec{x}$
- Nonlocal, based on entire functions of $\frac{\partial}{\partial t}$
 - Converge to sequence of ever-higher derivative theories
 - Virulently unstable if interacting
 - Contrary claims based on forbidding wild time dependence
 - If this is ok then classical GR has no singularities!

Why this is bad

- No guaranteed problem without interactions
 - Energy flow from $KE < 0$ to $KE > 0$ excites **both** DoF's
- No guaranteed problem without continuum DoF's
 - Instability is driven by vast d^3k UV phase space
 - Overwhelms even the weakest nonzero coupling
 - Decay is instantaneous
 - $\tau \neq 0$ results only come from imposing a UV cutoff
- Power and simplicity of the result
 - Requires only non-degenerate higher derivatives
 - Non-perturbative & independent of interactions
 - This is the strongest constraint on Fundamental Theory!

Decays happen because of entropy

- NOT to “lower its energy”
 - Energy is conserved!
 - $E(\text{excited}) = E(\text{ground} + \text{photon})$
- Quantum systems explore possibilities
- Many more ways to decay than not
 - Photon's energy is fixed
 - But it can go off in any direction
- $\pm$ Energy decays have even larger entropy
 - + Energy can be arbitrary
- NB must have 0-point motion to see this
 - Very safe assumption because $\hbar \neq 0$
 - But classical treatments must use minimum uncertainty initial value data

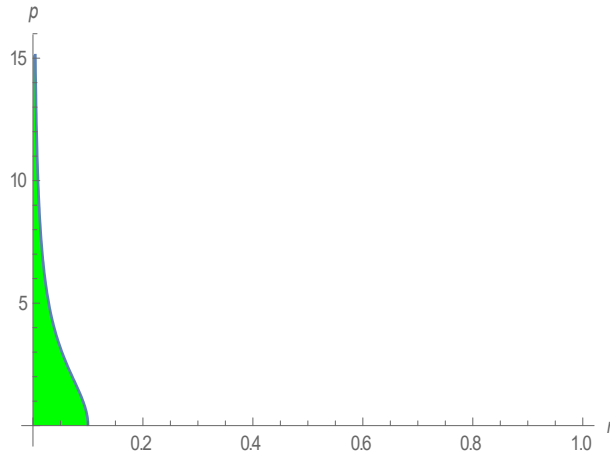


Common Misconceptions

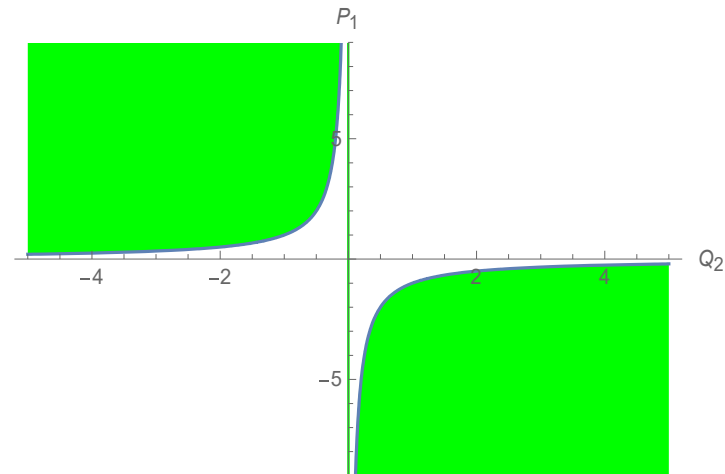
- “No problem for any $q(t) = q_0$ ”
 - Problem is pathological **time dependence** not special values
- “High mass ghosts decouple at low energies”
 - They actually couple more strongly!
- “No problem if HD’s confined to interactions”
 - Problem is non-perturbative
- “Quantization might help”
 - This is a large phase space problem

Quantization can prevent small phase space instabilities, not large ones

$$H = \frac{p^2}{2m} - \frac{e^2}{4\pi\epsilon_0 r} < -E$$



$$H = Q_2 P_1 < -E$$



Only hope is constraints (they compromise nondegeneracy)

- A single, global constraint on H does no good
- Problem is not that H decays
 - It's constant in fundamental theory!
 - Problem is vast # of states with $\pm$ energy DoF's giving same total energy
- Easy to work with gauge constraints
 - Just fix the gauge
 - Then ask: does the gauge-fixed theory contain nondegenerate HTD's?

The Case of $f(R)$ Gravity

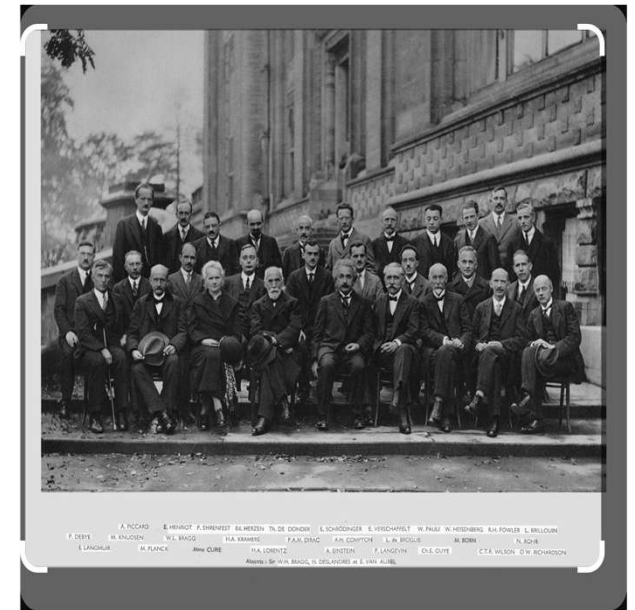
- HTD in $\mathcal{L}_{GR} = R\sqrt{-g}$ is degenerate (partially integrate)
- But $\mathcal{L} = f(R)\sqrt{-g}$ has a nondegenerate HD
 - But no ghosts
- NOT a violation of Ostrogradsky's theorem!
 - $P_1 Q_2$ unbounded below, but also ABOVE
 - Higher derivative DoF has opposite KE wrt lower derivative DoF
 - $R = (g^{\rho\mu} g^{\sigma\nu} - g^{\rho\sigma} g^{\mu\nu}) \partial_\mu \partial_\nu g_{\rho\sigma} + \text{lower derivatives}$
 - Only one metric component carries ∂_0^2
 - This component (Newtonian potential) is a ghost in GR
 - But no problem because constrained
 - So Ostrogradsky **predicts** new $f(R)$ DoF has positive KE

The Ugly

Literature discusses nonunitarity, not instability

- HTD's in particle theory don't have neg. energies
 - Particle theorists speak of negative norms
- They regard $-E$ annihilators as $+E$ creators
 - This involves nonnormalizable wavefunctions!
 - It also forfeits the classical Correspondence Limit!
- Then they try to decouple the neg. norm states
 - Nonunitarity from scattering into the $-$ states
- The various dodges can get VERY ugly
 - E.g., violations of microcausality
 - Sometimes they change the canonical norm

Quantum Mechanics (1900-13 & 1927-28)



$$L = -\frac{gm}{2\omega^2} \ddot{q}^2 + \frac{m}{2} \dot{q}^2 - \frac{m\omega^2}{2} q^2$$

- $q(t) = \sum [C_{\pm} \cos(k_{\pm}t) + S_{\pm} \sin(k_{\pm}t)]$
 - $k_{\pm} = \omega \sqrt{\frac{1 \mp \sqrt{1-4g}}{2g}}$ carries $\pm$ KE
 - $C_{\pm} = \frac{k_{\mp}^2 q_0 + \ddot{q}_0}{k_{\mp}^2 - k_{\pm}^2}$ $S_{\pm} = \frac{k_{\mp}^2 \dot{q}_0 + \ddot{q}_0}{k_{\pm}(k_{\mp}^2 - k_{\pm}^2)}$
- $H = \frac{m}{2} \sqrt{1-4g} [k_+^2 (C_+^2 + S_+^2) - k_-^2 (C_-^2 + S_-^2)]$
- $E > 0$ creation and annihilation operators
 - $\alpha \propto C_+ + iS_+$ $\alpha^{\dagger} \propto C_+ - iS_+$
- $E < 0$ creation and annihilation operators
 - $\beta \propto C_- - iS_-$ $\beta^{\dagger} \propto C_- + iS_-$

$$L = -\frac{gm}{2\omega^2} \ddot{q}^2 + \frac{m}{2} \dot{q}^2 - \frac{m\omega^2}{2} q^2$$

- Canonical positions

- $Q_1 \equiv q$ and $Q_2 \equiv \dot{q}$ recall $k_{\pm} = \omega \left[\frac{1 \mp \sqrt{1-4g}}{2g} \right]^{1/2} \rightarrow k_- - k_+ > 0$

- Positive norm, negative energy “empty” state

- $\Omega(Q_1, Q_2) \propto \text{Exp} \left[-\frac{mg}{2\hbar\omega^2} \left((k_- - k_+)k_+k_-Q_1^2 - 2ik_+k_-Q_1Q_2 + (k_- - k_+)Q_2^2 \right) \right]$

- Canonical quantization reproduces the Ostrogradskian instability

- Negative norm, positive energy ground state

- $\bar{\Omega}(Q_1, Q_2) \propto \text{Exp} \left[-\frac{mg}{2\hbar\omega^2} \left((k_- + k_+)k_+k_-Q_1^2 + 2ik_+k_-Q_1Q_2 - (k_- + k_+)Q_2^2 \right) \right]$

- This is non-normalizable nonsense

Normalizability puts the “quantum” in QM

(NB you can do this same trick for ANY theory)

This is Quantum Mechanics

- SHO $\rightarrow H = \frac{1}{2m} P^2 + \frac{1}{2} m \omega^2 Q^2$
- $a_{\mp} \equiv \sqrt{\frac{m\omega}{2\hbar}} \left[q \pm \frac{\hbar}{m\omega} \frac{\partial}{\partial q} \right]$
- $\Omega(q) \propto \exp \left[-\frac{m\omega}{2\hbar} q^2 \right]$
- $|N\rangle \propto (a_+)^N |\Omega\rangle$
- $H|N\rangle = +\left(N + \frac{1}{2}\right)\hbar\omega|N\rangle$
- Sum of Hermitian squares has **POSITIVE** spectrum

This is nonsense

- $H|\psi\rangle = E|\psi\rangle$ 2nd order ODE
 - 2 solutions for **every** E
- $\bar{\Omega}(q) \propto \exp \left[+\frac{m\omega}{2\hbar} q^2 \right]$
- $|\bar{N}\rangle \propto (a_-)^N |\bar{\Omega}\rangle$
- $H|\bar{N}\rangle = -\left(N + \frac{1}{2}\right)\hbar\omega|\bar{N}\rangle$
- Can get any other spectrum
 - Positive, negative, complex . .

Why not change the norm? (Alternate Quantization)

- Nothing wrong with alternate quantization for new fields
 - Physics an experimental science
 - Perhaps a new one requires alternate quantization
- But problematic for fields we understand classically
 - Taking $\hbar \rightarrow 0$ does not recover the known classical theory
- **Only** data from low energy gravity is classical GR
 - CMB perturbations come from high energy **normal** QGR
- $\lim_{\hbar \rightarrow 0} \overline{R + C^2}$ not even a local, metric theory
 - We have a complete catalog of these & it's not there

What's wrong with giving up classical Correspondence?

- Facetious Example: Define “Quantum Gravity” as SHO
 - Positive energy, positive norm, unitary, finite, completely solvable!
 - Of course it doesn't predict the tides . . .
 - Or the Solar System, gravitation radiation, black holes, cosmology . . .
- HDG people need to prove their model reproduces classical GR
- **IF** everything works (a BIG if) → START with this and forget HDG
 - Why torture us with HTD representation, alternate QM, acausality, etc.?
- Also cannot use classical HDG to study this model
 - Classical HDG is not the classical limit of the theory you claim to believe

Summary

- Phenomenological interest in HTD's due to Stelle's Theorem
 - Problem of QG is solved if the Weyl-squared term is ok
 - Of course some people just want to explore possible theories
- Ostrogradsky's theorem for nondegenerate HTD's
 - Hamiltonian linear in all but one canonical momentum
 - With interactions and continuum DoF → virulent kinetic instability
 - Simplest explanation for why Newton was right about $F = ma$!
- Particle theorists use negative-norm, positive E quantization
 - Regard negative E annihilators as positive E creators → non-normalizable states
 - Negative norm states violate probabilistic interpretation of QM
 - Excluding them violates unitarity because positive norms scatter into negative norms
 - Proposed fixes all sacrifice classical Correspondence Limit & problematic for gravity

I know most in this seminar series disagree (Witnessing to the heathen)

- I'll be here to (politely) debate Oct. 13 & 20
 - After that the seminar conflicts with teaching
- Don't waste your lives on a blind alley
 - There's so much to do in QG!
- My three favorite questions
 1. What caused primordial inflation?
 2. What caused current cosmic acceleration?
 3. What explains phenomena attributed to DM?
- Other things QG might do
 - Fluctuations of light-cone might permit starships!
 - Regulation of self-energy divergences

